Ifeanyi Desmond Dike

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Research Interests

Full-duplex spoken dialogue systems, turn-taking and overlap handling in voice AI, evaluation methodology for interactive speech agents, multimodal AI (speech-video-language), temporal sequence modeling, numerical methods and PDEs, financial mathematics.

Education

MSc in Financial Engineering 2026 – present
WorldQuant University
Coursework: Financial Data Science, Machine Learning for Finance, NLP for Finance, Financial Econometrics, Deep Learning, Stochastic Modelling, Portfolio Management, Risk Management

Bachelor of Technology in Applied Mathematics (First Class Honours) 2013 – 2018
Federal University of Technology Akure *Akure, Nigeria*
GPA: 4.63 / 5.0
Thesis: Couette-Poiseuille Reactive Fluid Flow with Variable Thermophysical Properties — coupled nonlinear PDEs, perturbation analysis, and finite difference numerics
Supervised by Prof. K.S. Adegbie

Preprints

Evaluating Overlap Handling in Gemini 3.1 Flash Live: A Reproduction of Full-Duplex-Bench v1.5
Ifeanyi Desmond Dike. *Zenodo*, May 2026.
doi:10.5281/zenodo.20354457

Reproducing Full-Duplex-Bench: Evaluation of Gemini 3.1 Flash Live on Turn-Taking and an Analysis of ASR Backend Sensitivity
Ifeanyi Desmond Dike. *Zenodo*, May 2026.
doi:10.5281/zenodo.20305268

Peer-Reviewed Publications

Fast Prediction of Equivalent Model of Installed Patch Antenna Radiation Pattern
H. Monday, J. Li, M. Raji, G. Nneji, **I. Dike**, R. Nneji.
IEEE IEMCON 2018, pp. 286–291.

An Improved e-Clearance Management System for Graduating Students in a University Environment
G. Nneji, J. Deng, S. S. Shakher, H. Monday, D. Agomuo, **I. Dike**.
IEEE IEMCON 2018, pp. 74–80.

Enhanced Attendance Management System: A Biometrics System of Identification Based on Fingerprint
H. Monday, **I. Dike**, J. Li, D. Agomuo, G. Nneji, A. Ogungbile.
IEEE IEMCON 2018, pp. 500–505.

Design of an Improved Cost Effective Electronic Locking System
H. Monday, J. Li, G. Nneji, C. Ukwuoma, **I. Dike**, R. Nneji.

IEEE IEMCON 2018, pp. 493–499.

Awards & Honors

Gold Medal — National Mathematics Competition for University Students (NAMCUS) 2017
Mathematical Association of Nigeria.

Bronze Medal — National Mathematics Competition for University Students (NAMCUS) 2016
Mathematical Association of Nigeria.

First Class Honours — Applied Mathematics, FUTA 2018
GPA 4.63/5.0.

Research Experience

Independent Researcher 2026 – present
Full-Duplex Spoken Dialogue Evaluation

- Reproduced Full-Duplex-Bench v1.0 and v1.5 (Lin et al., 2025) for Gemini 3.1 Flash Live Preview on Apple Silicon without CUDA
- Conducted first systematic analysis of ASR backend sensitivity in full-duplex evaluation — showed TOR varies by up to 62× across backends on identical audio
- Evaluated overlap handling across four scenarios; found systematic prosodic adaptation (higher pitch/WPM, lower intensity) after overlap events
- Two preprints published on Zenodo; arXiv submission in progress

Industry Experience

Founding AI Engineer — Datastrut AI, New York, U.S. (Remote) 2025 – present

- Built voice AI agents deployed in production for major U.S. construction contractors including Zachry Group, one of the largest privately held contractors in the U.S.
- Designed multi-caller feature enabling real-time AI-assisted collaboration between foremen and supervisors; closed enterprise deal with ICON Mechanical
- Conducted memory profiling and performance audits across all four microservices; identified and resolved CPU and memory bottlenecks that were costing thousands of dollars per month in cloud spend, reducing infrastructure costs by over 80%
- Migrated voice AI infrastructure from VAPI to self-hosted Pipecat on Fly.io

Co-Founder & Lead Developer — Cenphi, Remote 2025 – present

- Built multimodal pipeline jointly reasoning over speech, video, and language for testimonial intelligence
- Automated highlight extraction and sentiment scoring across 500+ hours of customer video; reduced manual review time from days to under an hour per campaign
- Speaker diarization, sentiment analysis, and highlight extraction at scale

Senior Software Engineer — Arbor Platform Inc, New York, U.S. (Remote) 2023 – 2024

- Integrated LLMs into video production workflows; 80% efficiency gain
- Built AI-assisted editing features that automated repetitive post-production tasks, reducing human review time across video content pipelines

Senior Software Development Engineer — Howdy Learning, New York, U.S. (Remote)

2020 – 2023

- Re-engineered WebSocket infrastructure; CPU usage reduced by 99%+
- Built real-time user presence system enabling live collaboration and activity tracking across community members; tracked video call progress and online status in real time
- Live video stream platform scaled to 10,000 concurrent connections

Technical Skills

ML & AI: PyTorch, HuggingFace Transformers, deep learning, LLM fine-tuning, RAG, vector databases, multimodal pipelines, computer vision
Speech & Voice: speech processing (Whisper, MLX, AssemblyAI), speaker diarization, voice AI (Pipecat), real-time audio streaming, evaluation methodology
Programming: Python, TypeScript, Golang, Rust, C++

References

Prof. K.S. Adegbie — Department of Mathematics, FUTA
Prof. B. Omolofe — Department of Mathematics, FUTA
Dr. S.O. Afolabi — Department of Mathematics, FUTA

Full contact details available on request.